Blessed G/Michael Catholic Secondary School

Semester 1 Physics Mid1 Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** According to the First Law of Thermodynamics, the change in internal energy (Delta U) is given by:

A. Delta U = Q - W **B.** Delta U = Q + W

C. Delta U = W - Q **D.** Delta U = Q / W

____ **2.** In an isothermal thermodynamic process, which physical variable remains constant?

A. Pressure **B.** Volume **C.** Temperature **D.** Entropy

____ **3.** What is the maximum theoretical efficiency of a Carnot heat engine operating between a hot reservoir at 600 K and a cold reservoir at 300 K?

A. 25% **B.** 33% **C.** 50% **D.** 75%

____ **4.** Two point charges of +2 microC and +8 microC are placed in vacuum separated by a distance r. If the distance is doubled (2r), the electrostatic force between them becomes:

A. Doubled **B.** Halved

C. One-fourth as large **D.** Four times larger

____ **5.** What is the electric field inside a hollow charged spherical conductor in electrostatic equilibrium?

A. Infinite **B.** Zero

C. Uniform and non-zero **D.** Dependent on radius

____ **6.** The electric potential V at a distance r from an isolated point charge Q in free space is given by:

A. k * Q / r **B.** k * Q / r^2 **C.** k * Q^2 / r **D.** k * Q * r

____ **7.** What is the capacitance of a parallel-plate capacitor with plate area A and separation distance d in air?

A. C = epsilon_0 * A / d **B.** C = epsilon_0 * d / A

C. C = A * d / epsilon_0 **D.** C = epsilon_0 * A * d

____ **8.** In an adiabatic thermodynamic process:

A. No heat enters or leaves the system (Q = 0)
B. Temperature is strictly constant
C. Pressure is zero
D. Work done is zero

____ **9.** The Second Law of Thermodynamics (Kelvin-Planck statement) asserts that:

A. It is impossible to construct a heat engine operating in a cycle that absorbs heat and converts 100% of it into work without rejecting waste heat
B. Heat can spontaneously flow from a colder body to a hotter body
C. Energy is not conserved in cycles
D. Absolute zero is achievable in finite steps

____ **10.** What is the change in entropy Delta S when 600 J of heat is added reversibly to an isothermal system at 300 K?

A. 2.0 J/K **B.** 0.5 J/K **C.** 180,000 J/K **D.** 120 J/K

____ **11.** According to Gauss's Law, the total electric flux through any closed Gaussian surface is equal to:

A. Q_enclosed / epsilon_0 **B.** Q_enclosed * epsilon_0

C. Zero always **D.** E * Area

____ **12.** Electric field lines always originate on:

A. Positive charges and terminate on negative charges
B. Negative charges and terminate on positive charges
C. Closed continuous loops only
D. Equipotential surfaces only

____ **13.** The relation between electric field E and electric potential V in one dimension is:

A. E = -dV / dx **B.** E = dV / dx **C.** E = V * x **D.** E = -V / x^2

____ **14.** What is the work required to move a +5 microC charge across an equipotential surface between two points with potential difference Delta V = 0?

A. 0 Joules **B.** 5 Joules **C.** 25 Joules **D.** Undefined

____ **15.** Inserting a dielectric material with constant k > 1 between the plates of an isolated charged capacitor:

A. Increases capacitance and decreases electric potential difference
B. Decreases capacitance
C. Leaves capacitance unchanged
D. Reduces charge on plates to zero

____ **16.** The electric energy U stored in a capacitor with capacitance C charged to voltage V is:

A. (1/2) * C * V^2 **B.** C * V

C. (1/2) * C^2 * V **D.** Q / V

____ **17.** Two capacitors of 6 microF and 3 microF are connected in series. What is the equivalent capacitance?

A. 2 microF **B.** 9 microF **C.** 0.5 microF **D.** 18 microF

____ **18.** What is the electric field at distance r outside an infinitely long straight wire with linear charge density lambda?

A. lambda / (2 * pi * epsilon_0 * r) **B.** lambda / (4 * pi * epsilon_0 * r^2)

C. lambda / epsilon_0 **D.** Zero

____ **19.** An ideal gas expands from 2 m^3 to 5 m^3 at constant pressure of 200 kPa. What is the work done by the gas?

A. 600 kJ **B.** 400 kJ **C.** 1,000 kJ **D.** 100 kJ

____ **20.** The Zeroth Law of Thermodynamics provides the operational definition of which physical quantity?

A. Temperature **B.** Entropy **C.** Work **D.** Heat capacity

____ **21.** What is the root-mean-square (rms) speed of ideal gas molecules of molar mass M at absolute temperature T?

A. sqrt(3RT / M) **B.** sqrt(2RT / M) **C.** 3RT / (2M) **D.** sqrt(RT / 3M)

____ **22.** An electric dipole placed in a uniform electric field experiences:

A. A net torque but zero net force **B.** Both net force and net torque

C. A net force but zero torque **D.** Neither force nor torque

____ **23.** What is the dimension of electric potential (Volt)?

A. [M L^2 T^(-3) I^(-1)] **B.** [M L T^(-2) I^(-1)]

C. [M L^2 T^(-2)] **D.** [M L^3 T^(-3) I]

____ **24.** In an isobaric expansion of an ideal gas:

A. Pressure remains constant while volume and temperature change
B. Volume remains constant
C. Internal energy is zero
D. No work is done

____ **25.** What is the electric field magnitude at a distance of 3 m from a point charge of 9 nC in vacuum? (k = 9 * 10^9 N*m^2/C^2)

A. 9 N/C **B.** 1 N/C **C.** 27 N/C **D.** 81 N/C